

Muckund Sharma

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Mechanical Engineering

GPA: 3.98/4.0

Relevant Coursework: Mechanics, Material Structure and Properties, Control of Material Properties

Waterloo, ON

Sep. 2025 – Apr. 2030

EXPERIENCE

Mechanical Design Team Member

Sep. 2026 – Present

Waterloo Aerial Robotics Group (WARG)

Waterloo, ON

- Designed a lightweight landing-gear subassembly using 22 mm OD carbon-fiber tubes and a 3D-printed PETG-CF bracket, incorporating an intentional breakaway section to protect the primary airframe during hard crashes.
- Validated landing-gear bracket performance using orthotropic linear-static FEA under 1–4g loading, achieving a 3.81 FOS at 1g while predicting tensile-failure onset at 4g to enable sacrificial failure.
- Refined the FEA mesh around bolt interfaces and the breakaway region, observing only a 1.22% change in peak maximum principal stress (37.33 to 37.79 MPa) across refinement, confirming mesh convergence.
- Designed a 3D-printable protective ESC enclosure around an existing APD 120F3 ESC, incorporating M3 mounting interfaces, component clearances, passive ventilation, and terminal protection.

Core Airframe Team Member

Sep. 2025 – May 2026

Waterloo Rocketry

Waterloo, ON

- Performed compression testing on six fiberglass laminate configurations for the Polaris rocket airframe, following ASTM D695 / DIN EN 2850 procedures using a 15 kN-capacity testing system to characterize performance.
- Developed a Python analysis pipeline to automate processing of load/displacement data, generating stress-strain curves and extracting key failure properties for quantitative laminate comparison.
- Machined a Boeing BSS 7260 compression-testing fixture from technical drawings using manual milling and lathe operations, maintaining dimensional tolerances to ensure consistent coupon alignment and test accuracy.

Project Coordinator & Teaching Assistant

May 2026 – Aug. 2026

York Region Educational Services (YRES)

Markham, ON

- Mentored and evaluated 6 summer program staff, conducting daily check-ins, and weekly one-on-one feedback through a formal mentorship program, driving measurable performance growth across the team.
- Built and launched a project-tracking workspace in Jira for a large annual volunteer event, breaking the event plan into tasks and subtasks, organizing them into To Do/In Progress/Done stages, and assigning ownership, giving the team real-time visibility into progress for the first time.

PROJECTS

RC Aircraft Design, CFD Analysis and Fabrication

Aug. 2026 – Present

- Designed and fabricated a complete Cessna 172-inspired RC aircraft, developing the airframe and engineering drawings in SolidWorks and translating the CAD design into a hand-built functional prototype.
- Performed 3D CFD analysis of the wing geometry in ANSYS Fluent, evaluating aerodynamic performance across multiple angles of attack and extracting lift and drag forces to calculate C_L , C_D , and L/D .
- Used CFD results to guide wing-incidence selection, identifying 2.26° as the most efficient configuration within the evaluated design space, with a maximum simulated $L/D \approx 13.9$ and $C_L = 1.294$.
- Analyzed pressure, velocity, turbulence, and wall-shear distributions to characterize the wing's aerodynamic behavior and interpret sources of lift and drag.
- Integrated the propulsion and flight-control systems by selecting and installing a 1400-kV brushless motor, 30-A ESC, propeller, servos, receiver, and control surfaces into the fabricated airframe.

TECHNICAL SKILLS

CAD & Simulation: SolidWorks (CSWA Certified), AutoCAD, Ansys Mechanical/Fluent, CATIA, FEA, CFD

Programming & Data: Python, MATLAB, C++, Java, Git

Manufacturing & Fabrication: CNC Machining, Milling, Lathe, 3D printing, Laser Cutting, GD&T, Soldering

Electronics: Arduino, RC Systems, Brushless Motors/ESCs, Servos